

Gayuru Ramanayake

ELECTRICAL ENGINEERING UNDERGRADUATE

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Summary

An enthusiastic final-year **Electrical Engineering undergraduate** at the **University of Moratuwa** with strong interests in artificial intelligence, robotics, automation, and data-driven systems. Recently completed a **6-month AI/ML internship** at **Jaseci Labs** contributing to cutting-edge projects in vision-language models, and AI-driven applications. Currently pursuing **CIMA** (Chartered Institute of Management Accountants) to strengthen knowledge in business analytics, finance, and strategic management. Passionate about applying technology and analytical thinking to solve complex engineering and business challenges.

Education

Electrical Engineering

University of Moratuwa

Undergraduate, B.Sc. in Engineering (Hons) | Electrical Engineering | Control system and Automation

CGPA: 3.69 out of 4.00 (up to semester 7)

Dean's List: Semester 4 (3.90 / 4.00), Semester 6 (4.0/4.0)

Moratuwa, Sri Lanka

2022 - present

CIMA (Chartered Institute of Management Accountants)

Reading for CIMA Strategic Level

CIMA Advanced Diploma in Management Accounting (CIMA Adv Dip MA) (Management Level completed)

CIMA Diploma in management accounting (Operational Level completed)

Certificate Level Completed

present

GCE Advanced Level Examination

BANDARANAYAKE COLLEGE GAMPAHA

Mathematics - A, Physics - A, Chemistry - A, (z- score - 2.2172)

Sri Lanka

2020

GCE Ordinary Level Examination

BANDARANAYAKE COLLEGE GAMPAHA

9As (A passes for all 9 subjects)

Sri Lanka

2017

Experience

AI/ML Engineering Intern – Jaseci Labs

Dec 2024 – June 2025

Technologies: Jaclang, Python, Docker, Kubernetes, gRPC, FastAPI, React, Redux, Next.js, Hugging Face, Hydra, W&B

Key Contributions:

- **Backend Re-Architecture:** Rebuilt the complete backend for MyCA AI, an AI-powered productivity tool for high-performing individuals, using Jaclang 2.0 while co-authoring the system architecture and technical API documentation.
- **VLM Fine-Tuning:** Fine-tuned Vision-Language Models (VLMs) on custom datasets (Car Damage, RSICD) using LoRA/QLoRA and Unsloth optimization techniques for efficient training.
- **Frontend Development:** Developed the frontend interface for a Small Language Model (SLAM) evaluation tool using React, Redux, and Next.js.
- **DevOps & Deployment:** Containerized microservices using Docker and deployed gRPC APIs on Kubernetes clusters with active Load Balancing.
- Gained hands-on experience with AI model evaluation, fine-tuning, and productization in a collaborative global environment.

Projects

• Vision-Language-Model Powered Intelligent Assistance System

University of Moratuwa

FINAL YEAR RESEARCH PROJECT (ONGOING PROJECT)

2025-present

Technologies: Python, Vision-Language Models (VLMs), NVIDIA Jetson, Speech-to-Text/Text-to-Speech, PyTorch, Hugging Face

- **System Development:** Engineering a real-time assistive navigation system for the visually impaired using Vision-Language Models (VLMs) for semantic scene understanding and obstacle awareness.
- **Edge Optimization:** Fine-tuning VLMs and implementing a selective attention mechanism on NVIDIA Jetson edge hardware to achieve low-latency inference for critical environmental regions.
- **Open-Source Impact:** Curated and published the *blind-assist/walk-train* dataset on Hugging Face, which has secured **14,800+ community downloads**, becoming a globally recognized resource for assistive robotics research.

• Autonomous Maze Solving Robot – 1st Place, Maze Solver Challenge 2025

University of Moratuwa

EMBEDDED SYSTEMS DESIGN PROJECT (MAZE SOLVER CHALLENGE PROJECT 2025)

2025

Technologies: C++, Arduino Mega, Flood Fill, BFS, PID Control, Sensor integration (ultrasonic, IR)

- Engineered a hybrid Flood Fill + BFS navigation system with dual-mode PID control for wall-following and precision line-following across two isolated maze sections.
- Designed a custom 3D-printed chassis with real-time sensor fusion for seamless mode switching.

• Smart Fire detection and mitigation system

University of Moratuwa

MEASUREMENT PROJECT

2024

Technologies: Python, OpenCV, Haar Cascade, LabVIEW, TCP/IP, Flame/Temperature/Gas Sensors

- Built a real-time fire detection system combining camera-based CV and multi-sensor fusion (flame, temperature, gas) with automated actuator control.

- Established TCP/IP bridge between Python and LabVIEW for live monitoring and actuation.
- **Auto-Tracking Security Camera System** University of Moratuwa
2023
DESIGN PROJECT
Technologies Used: MATLAB, SolidWorks, SimScape, and Robotics System Toolbox
 - Designed camera mechanical components in SolidWorks and simulated autonomous pan-tilt tracking behavior using SimScape and Robotics System Toolbox.
 - Implemented real-time object detection and tracking logic in MATLAB for autonomous surveillance of moving subjects.
- **Smart Medibox: Automated Light-Responsive Medicine Storage System** University of Moratuwa
2024
EMBEDDED SYSTEM INDIVIDUAL PROJECT
Technologies Used: Arduino, Node-RED, MQTT
Developed an intelligent medicine box that adjusts a sliding window based on ambient light conditions using a light sensor and a servo motor. The system includes a real-time monitoring interface and customizable control through a Node-RED dashboard.
 - Designed and implemented a control system for automated sliding window adjustment based on light intensity.
 - Developed a real-time monitoring and control interface using Node-RED with customizable thresholds.
- **Logistic Vehicle Machinery Services and Maintenance Subsystem** University of Moratuwa
2024
MODULAR SOFTWARE DEVELOPMENT PROJECT
Company A | Scrum Master & Backend Developer
Technologies: Jira, Java (Spring Boot), React, API Development
 - Led the Agile/Scrum process on Jira and engineered the backend REST APIs using Spring Boot for a fleet maintenance system.
 - Integrated the backend with a React frontend to streamline vehicle service scheduling and tracking operations.
- **Potato Disease Detection System**
INDIVIDUAL PROJECT
Technologies: Deep Learning, TensorFlow, CNN, FastAPI, React
 - Developed a Convolutional Neural Network (CNN) using TensorFlow to classify plant diseases, utilizing data augmentation to improve model generalization.
 - Exported the trained model locally for further use.

Publications

- G. Ramanayake et al., "Systematic Benchmarking and Edge-Aware Selection of Compact Vision-Language Models for Blind and Low-Vision Navigation Assistance," 2026 IEEE 7th World AI IoT Congress (AIIoT 2026), Seattle, USA, Accepted for publication (camera-ready under preparation).

Volunteering

- EESOC Member (Electrical engineering society) - PR team member University Of Moratuwa
- Maths society - Logistic Pillar Member - ENIGMA - Maths Society (2024)) University Of Moratuwa
- Company Coordination – EESPIRE (Career Day Electrical engineering department) 2024 University Of Moratuwa

Skills

- **Simulation** - MATLAB, PSCAD
- **Programming** - C/C++, Python, JAVA, Lab view, HTML, CSS, JavaScript, Arduino, Jaclang
- **Backend** – Spring boot
- **DBMS** – MongoDB
- **Design** - MS Office Package, Figma, Canva
- **Modeling** - Simulink, Solidworks, easycad
- **AI/ML Tools** - Hugging Face, LoRA, QLoRA, Hydra,
- **DevOps** - Docker, Kubernetes, RunPod
- **API & Microservices** - FastAPI, gRPC
- **Others** - Git/ GitHub, Matplotlib, NumPy, Weights & Biases
- **Soft Skills** - Problem solving and Critical thinking, leadership, Teamwork, Time management, Adaptability

Certifications

- Machine Learning** Stanford University
- Intro to Machine Learning** Kaggle
- Python for Beginners** Open Learning Platform - UoM
Verification code -HHQO9XshHA
- Certificate in Business Accounting** CIMA
Credential ID 1-6AIHZSZ

Languages

- English
- Sinhala Native

References

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